SolarVision Mower: Autonomous Solar-Powered Lawn Maintenance System

Team Number: 20

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ECE 414 – Senior Design Project

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Abstract

Combines monocrystalline solar panels, a Li-ion battery, and an ML-based path planning system.

LiDAR, GPS, and Camera-based fully autonomous navigation.

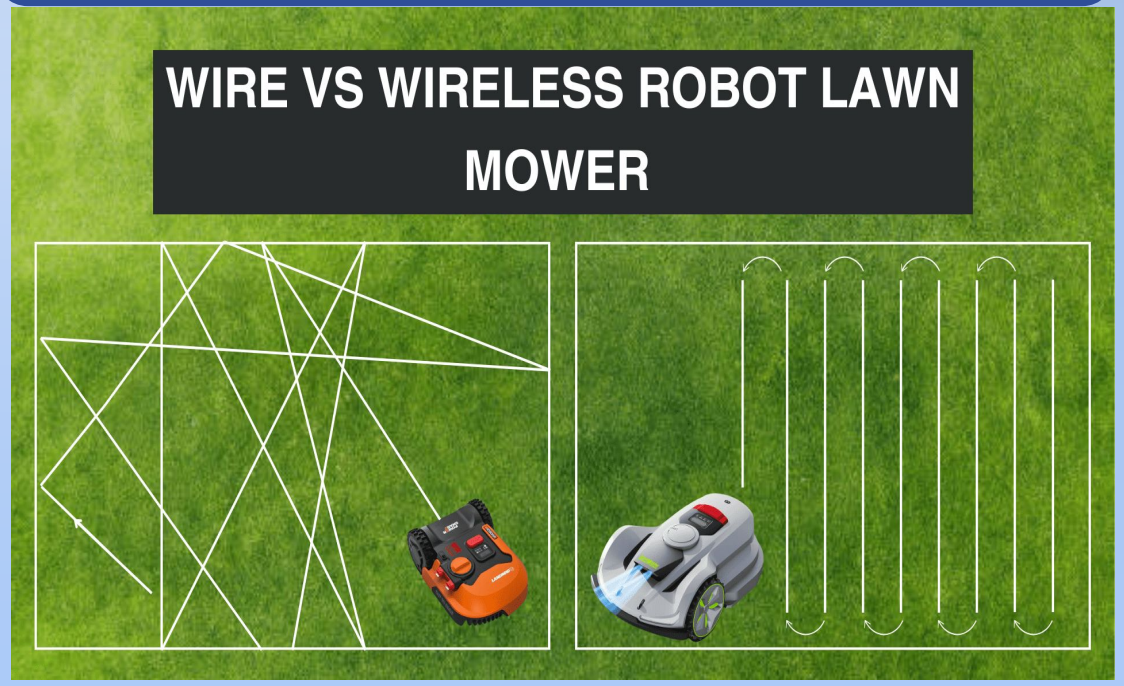
Objectives of minimizing energy use and getting rid of fossil-based reliance.

The main components: STM32N6 MCU, DC motor controllers, LiDAR, GPS module, Camera, and MPPT solar charger.

Caters to residential and small commercial customers with the need to have eco-efficient lawn care.

Combines robotics, edge computing, IoT, and renewable energy technologies [1], [2].

Evolution of robotic lawnmowers from wired navigation to sensor-based intelligent systems [9].

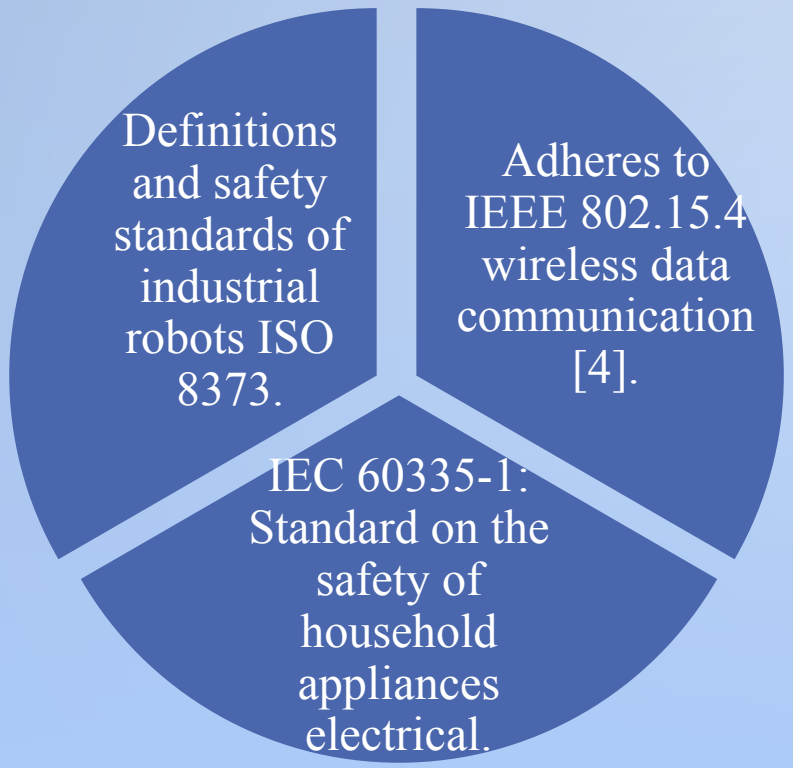


Part 1 Design Objectives, Specifications, and Constraints.

Objective 1:	• To create a completely independent solar-powered mower that can work throughout the day.
Objective 2:	• LiDAR, GPS, and Camera based SLAM obstacle avoidance and path planning
Objective 3:	• Mobile/Web interface to monitor mower's status and operation
Power supply:	• 3 12V monocrystalline solar panels, 36 V 10Ah Li-ion battery (10S 4P config).
Hours of average run-time:	• ~1.5 hours on full charge, assuming 200W power consumption
Safety limitations:	• Enclosure, slow control (less than 2 mph), boundary enforcement, and tilt detection.
Budget constraint:	• Prototype should not go above $600.

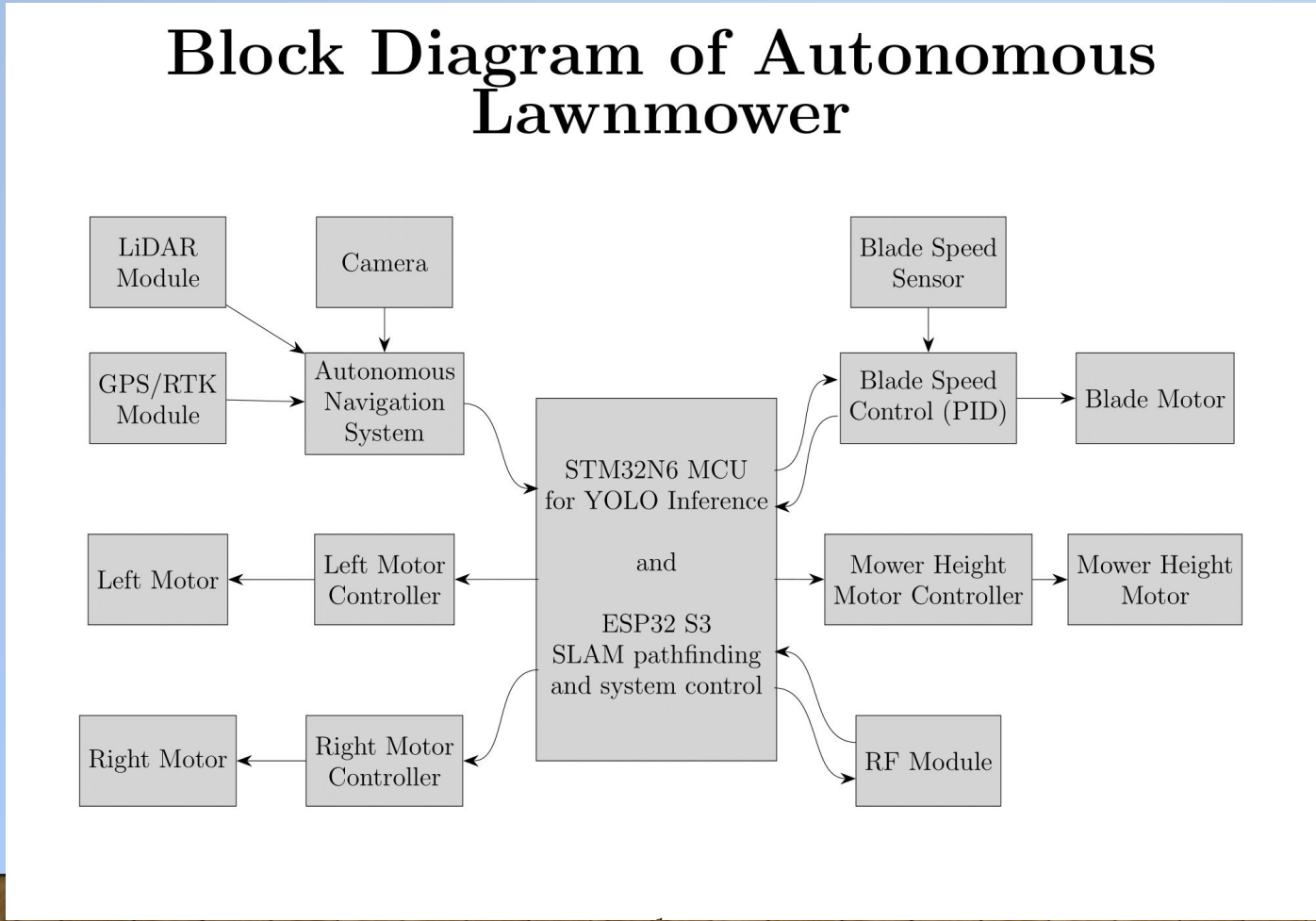
Proposed SolarVision Mower and its docking deck models

Part 2 Standards and Technical Constraints



Safety constraint:	Implement tilt sensor, disable blade upon being lifted
Production limitation:	Modular design for efficient and practical assembly
Space constraints:	Mower will take up no more space than standard.
Boundary Enforcement:	Enforce strict boundaries in path finding algorithm

Blocks and Components Diagram



Microcontroller:
STM32N6 and Raspberry Pi

Sensors:
RPLiDAR C1, Adafruit GPS module, and camera.

Subsystem of power:
MPPT charge controller, 3 12V solar panels, and 10S3P Li-ion battery pack.

Motor system:
Two 12V DC propulsion motors, one cutting blade motor.

Communication:
Wi-Fi module having a web dashb-oord interface.

Prototyping, Testing, Assembly, and Validation.

Prototyping:

- Primary Type of Prototyping:

Functional Prototyping

- To make sure the core systems actually work even if the final form is imperfect.

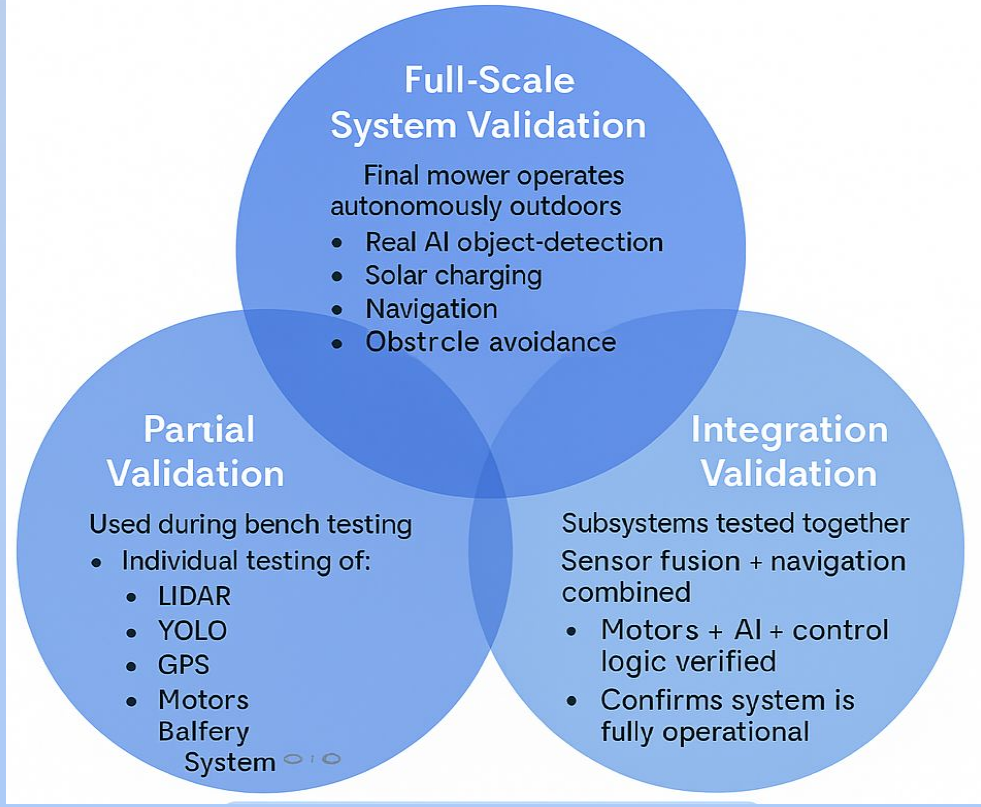
- Final Type of Prototyping:

Working Prototype

- To make sure its a fully operational mower

- Earlier Phases of Prototyping:

Proof of Concept Prototyping



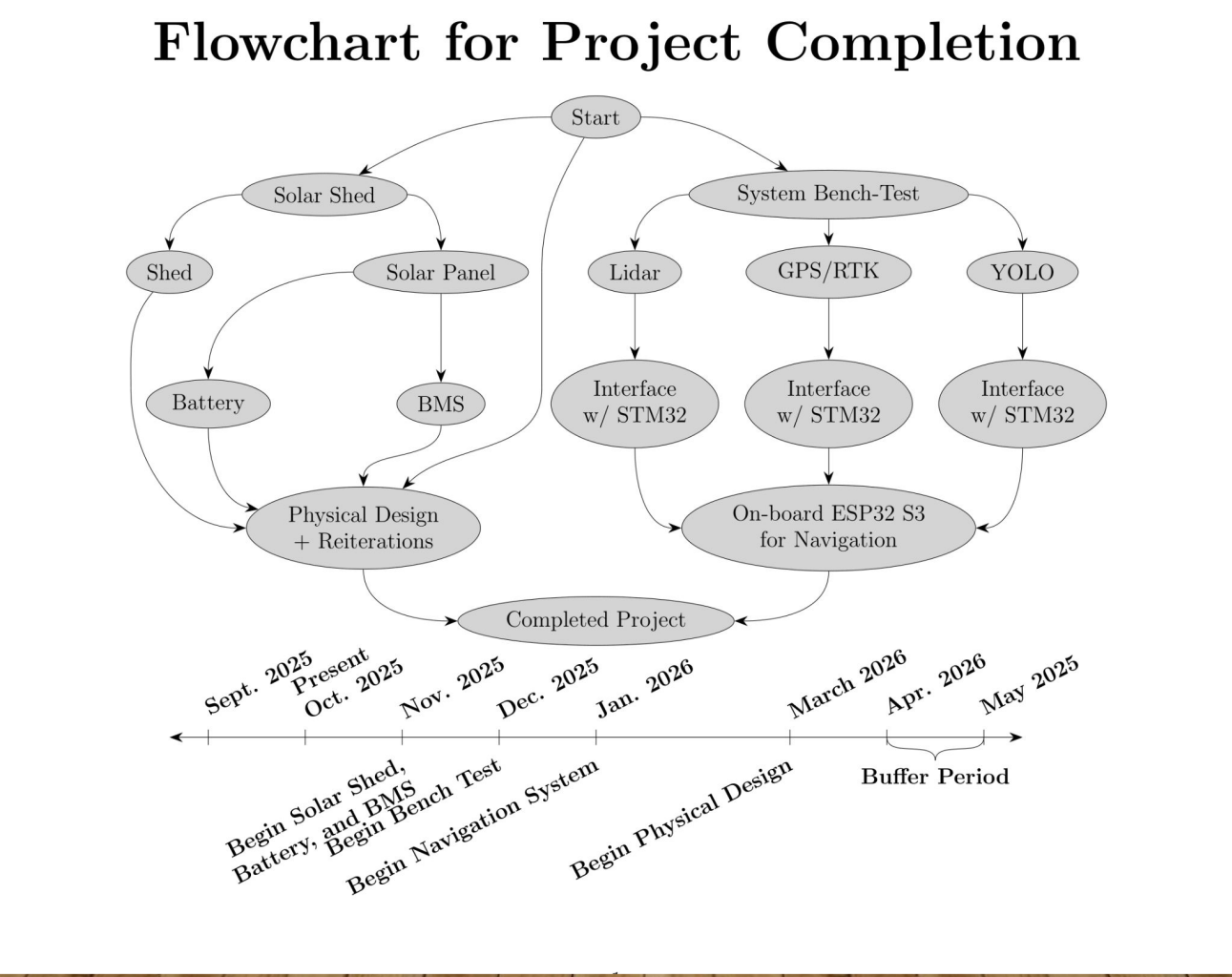
Testing Category

Bench Testing → Integration Testing → Field & Validation Testing

Assembly Category

Subsystem Assembly → Full System Assembly

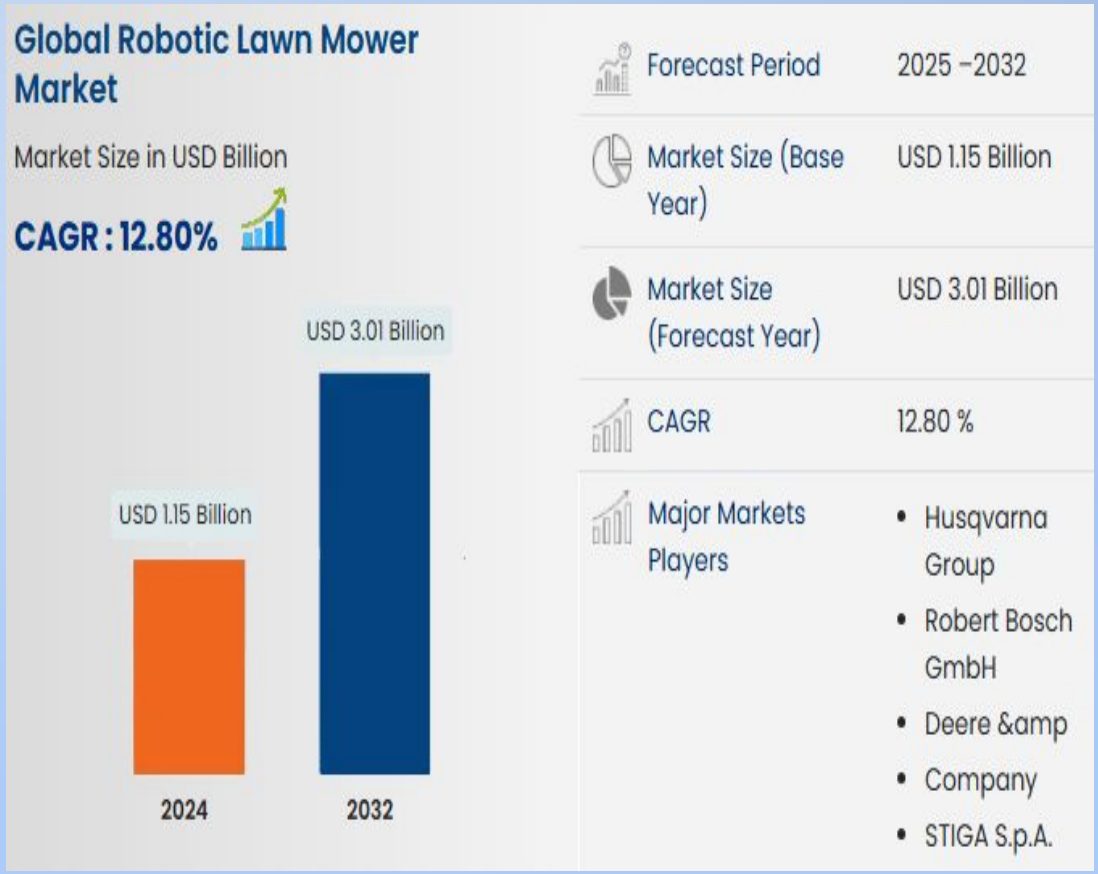
Prototyping, Testing, Assembly, and Validation (Cont.)



Business, Economic Potential, and Benchmarking

Global robotic lawn mower market trends [17].

- Rivals Husqvarna Automower, WORX Landroid.
- Novelty: artificial intelligence and AI-assisted navigation [6].
- Retail cost expected: around 850 dollars (compared to an average of 1,200 for competitors)
- Market niche: residential users who are green-friendly.
- Potential of the economy: lowering the energy cost, no charge dock needed.
- The payback period is estimated to be 2 years after saving energy.



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